

SONU YADAV, Ph.D.

Experimental Plasma Physics | Advanced Plasma Diagnostics | Scientific/Hardware Engineering | Research Leadership

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PROFESSIONAL PROFILE

Experimental plasma physicist with 10+ years of research and engineering experience across magnetized plasmas, plasma diagnostics, RF plasma sources, plasma-surface interactions, and scientific instrumentation. Experienced in leading complex experimental campaigns, developing laser- and probe-based diagnostics, operating high-vacuum plasma facilities, and integrating RF power, electronics, automation, and data analysis. Research spans magnetic reconnection, turbulence and wave-particle interactions, advanced plasma diagnostics, semiconductor plasma processing, and electric propulsion. Strong record of international collaboration, peer-reviewed research, international collaboration, invited presentations, technical leadership, mentoring, and cross-functional experimental development.

CORE EXPERTISE

Plasma Diagnostics

Thomson scattering, laser-induced fluorescence (LIF), OES, Langmuir/Mach probes, B-dot arrays, Faraday cups, RFEA

Vacuum & Experimental Systems

UHV/HV chambers, feedthroughs, gas delivery, leak diagnostics, RF shielding/grounding, experimental integration

Software & Analysis

MATLAB, Python, LabVIEW/LabWindows, Mathematica, Origin; signal processing, DOE, sensitivity analysis

RF & Plasma Sources

Helicon, ICP, CCP, pulsed RF systems, plasma guns, hot-cathode DC discharges, ion-beam systems

Engineering & Instrumentation

SolidWorks, CATIA, KiCad, LTSpice, PCB/signal-chain design, motorized probe drives, low-noise acquisition

Research Domains

Magnetic reconnection, turbulence, wave-particle interactions, plasma instabilities, optical plasma diagnostics, RIE/ALE, expanding plasmas, electric propulsion

RESEARCH LEADERSHIP, FUNDING & PROFESSIONAL SERVICE

Researcher III | Department of Physics & Astronomy, West Virginia University

May 2022 – Present

Morgantown, West Virginia, USA

- Lead experimental operations and research campaigns on the PHase Space Mapping (PHASMA) facility for collisionless magnetic reconnection, plasma turbulence, energy dissipation, and particle energization.
- Lead, maintain, and integrate pulsed plasma guns, vacuum systems, timing/trigger systems, data acquisition, electric and magnetic probe arrays, and Thomson scattering diagnostics, Triple Langmuir probe array.
- Develop and deploy high-resolution plasma diagnostics for electron and ion velocity/energy distribution measurements, including Thomson scattering, Langmuir probes, B-dot arrays, RFEAs, and optical diagnostics.

- Investigate guide-field reconnection, lower-hybrid drift waves, wave-particle interactions, electron energization, and MHD-to-kinetic-scale turbulence using synchronized multi-diagnostic measurements.
- Lead an internally funded research project on plasma atomic layer etching of MoS₂ for next-generation semiconductor devices; contribute to CCP/RIE source operation with inert and reactive gases.
- Mentor graduate researchers, support collaborative user experiments, contribute to proposal development, and coordinate experimental planning across multidisciplinary teams.

Postdoctoral Fellow | Institute for Plasma Research (Homi Bhabha National Institute)

May 2020 – May 2022

Gandhinagar, India

- Lead development of a 5-kW RF helicon plasma thruster for electric propulsion, targeting approximately 100 mN thrust.
- Designed and operated magnetized expanding plasma columns, magnetic nozzles, and ICP/helicon plasma sources for propulsion-relevant experiments.
- Developed expertise in high-power RF delivery and matching, vacuum-compatible hardware, plasma source optimization, diagnostics, and plasma-surface interaction measurements.

SELECTED TECHNICAL ACHIEVEMENTS

- RF plasma engineering: designed and operated high-density ICP, CCP, and helicon sources; implemented RF matching and tuning at 13.56 MHz and variable-frequency operation up to 2.5 kW.
- Diagnostic development: built and deployed Thomson scattering, multi-channel B-dot arrays, Langmuir and Mach probes, RFEAs, Faraday cups, LIF, fast-photodiode, and high-speed imaging diagnostics.
- Automation and data systems: integrated LabVIEW/LabWindows control, synchronized acquisition, motorized probe positioning, and MATLAB/Python analysis for repeatable experimental campaigns.
- Mechanical/electrical integration: designed experimental hardware and diagnostics using SolidWorks/CATIA; developed PCB layouts and low-noise signal conditioning using KiCad and LTSpice.
- Vacuum systems: operated and maintained systems spanning approximately 10^{-8} to 10^{-4} Torr, including chambers, feedthroughs, gauges, gas delivery, and troubleshooting.

EDUCATION

Ph.D. in Plasma Physics | Institute for Plasma Research / Homi Bhabha National Institute

2020

Gandhinagar, India

- Thesis: Effect of inhomogeneous magnetic field on helicon antenna produced expanding plasma. Supervisor: Prof. Prabal K. Chattopadhyay.

M.Sc. in Physics (Material Science) | Devi Ahilya Vishwavidyalaya

2012

Indore, Madhya Pradesh, India

- Project: Preparation and Characterization of Graphene using Raman Spectrometer.

- Major areas: Computer Science and Physics.

LEADERSHIP, AWARDS & PROFESSIONAL SERVICE

- **Principal Investigator**, WVU Internal Seed Funding (approved 2025): Plasma Atomic Layer Etching (ALE) of Molybdenum Disulfide (MoS₂) for Next-Generation Semiconductor Devices.
- **Chair, MagNetUS** User Working Group; Executive Committee Member, MagNetUS.
- **Session Chair** at APS Division of Plasma Physics (APS-DPP) and IEEE International Conference on Plasma Science (ICOPS).
- **Peer reviewer** for Physics of Plasmas, Physics of Fluids, Review of Scientific Instruments, and The Astrophysical Journal.
- **Poster judge**: AAS 248 Chambliss Poster Competition (2026) and LCSEE Graduate Student Poster Competition, West Virginia University (2026).
- **DOE proposal concept**: Experimental Investigation of Multiple X-Points During Electron-Only Reconnection; full submission encouraged (funding under review in source CV).
- **Lead Proposal**: High-Beta Expanding Plasma Experiments at the Large Plasma Device (LAPD), UCLA — Lead proposer; white paper submitted and currently under review (2026).

SELECTED PUBLICATIONS

- **Sonu Yadav**, T Rood, Earl Scime et al. A pathway for electron energization in strong Guide field magnetic reconnection, Published in **Physical Review Letter (2026)**
- G. E. Bartolo, T. Rood, **Sonu Yadav**, E. Scime, “Multi-Tip Floating Potential Probe with Active Compensation for Fluctuation Measurements,” Review of Scientific Instruments (2026).
- R. Nirwan, **Sonu Yadav**, K. J. Stevenson, G. E. Bartolo, K. Kumar, P. A. Cassak, E. Scime, “Guide-field-dependent electron acceleration during electron-only magnetic reconnection in the PHASMA experiment,” Physics of Plasmas 32, 122115 (2025).
- R. Nirwan, **Sonu Yadav**, T. E. Steinberger, E. Scime, “Designing an energy analyzer for measuring non-thermal electron energy distribution functions in the PHASMA experiment,” Review of Scientific Instruments 96, 123501 (2025).
- T. Rood, **Sonu Yadav**, G. Bartolo, E. Scime, “Fast photodiode arrays for high frequency fluctuation measurements of reconnecting flux ropes,” Review of Scientific Instruments 95, 093525 (2024).
- G. Bartolo, **Sonu Yadav**, C. Fitz, E. Scime, “Coherent Mode and Turbulence Measurements with a Fast Camera,” Review of Scientific Instruments 95, 093511 (2024).
- **Sonu Yadav**, S. Ghosh, S. Bose, K. K. Barada, R. Pal, P. K. Chattopadhyay, “Role of ion magnetization in formation of radial density profile in magnetically expanding plasma produced by helicon antenna,” Physics of Plasmas 25, 043518 (2018).
- **Sonu Yadav**, K. K. Barada, S. Ghosh, J. Ghosh, P. K. Chattopadhyay, “Effect of inhomogeneous magnetic field on plasma generation in a low magnetic field helicon discharge,” Physics of Plasmas 26, 082109 (2019).
- S. Ghosh, **Sonu Yadav**, K. K. Barada, P. K. Chattopadhyay, J. Ghosh, R. Pal, D. Bora, “Formation of annular plasma downstream by magnetic aperture in the helicon experiment,” Physics of Plasmas 24, 020703 (2017).

MANUSCRIPTS IN REVIEW

- Earl Scime, Sonu Yadav et al., “Exploration of Kinetic Plasma Physics Through In-situ Phase-space Measurements in Laboratory Plasmas,” Physics of Plasmas (2026).

SELECTED INVITED PRESENTATIONS

- “Time-Resolved Kinetic Diagnostics of Wave–Particle Interactions in PHASMA,” 248th American Astronomical Society Meeting, Pasadena, California, USA (June 2026).
- “Measuring Wave-Driven Electron Energization: Diagnostic Insights from the PHASMA Experiment,” IEEE ICOPS, Lake Tahoe, Nevada, USA (June 2026).
- “Direct Measurements of Electron Energization During Magnetic Reconnection,” 68th APS Division of Plasma Physics Meeting, Chicago, Illinois, USA (November 2026).

PROFESSIONAL MEMBERSHIPS

• American Physical Society • IEEE • European Physical Society • Plasma Science Society of India (member since 2013)

ADDITIONAL INFORMATION

- Active contributor to the international plasma physics community, with 20+ conference and workshop presentations/participations across North America, Europe, and Asia, including multiple invited presentations.
- Languages: English and Hindi.
- International research experience: United States and India; extensive participation in international scientific collaborations and conferences.
- References available upon request.